

# Homework 2

## Question 1.

Let  $f \in C^2([a, b])$ ,  $f(a)f(b) < 0$ ,  $f'$  and  $f''$  do not vanish and do not change signs on  $[a, b]$ .

- (a) Show that  $f(x) = 0$  has a unique solution  $x_* \in (a, b)$ .
- (b) Draw pictures to illustrate four different cases:
  - $f' > 0, f'' < 0$
  - $f' < 0, f'' > 0$
  - $f' > 0, f'' > 0$
  - $f' < 0, f'' < 0$
- i. Prove that if  $f'$  and  $f''$  have opposite signs, then Newton's Method starting at  $x_0 = a$  must converge to  $x_*$ .
- ii. Now suppose that  $f'$  and  $f''$  have the same sign. Which starting point should we choose to guarantee convergence?
- (c) Let  $M = \max \{|f''(x)| : x \in [a, b]\}$  and  $m = \min \{|f'(x)| : x \in [a, b]\}$ . Show that the errors  $e_n = x_n - x_*, n = 0, 1, 2, \dots$ , satisfy

$$|e_{n+1}| \leq \frac{M}{2m} |e_n|^2$$

## Question 2.

- (a) Let  $g \in C^1([a, b])$ . Suppose  $g(x) \in [a, b]$  and  $|g'(x)| < 1$  for all  $x \in [a, b]$ . Show that (i)  $g(x) = x$  has a unique solution  $x_* \in [a, b]$ , (ii) the iteration  $x_{n+1} = g(x_n)$  must converge for any  $x_0 \in [a, b]$ , and (iii)  $\lim_{n \rightarrow \infty} x_n = x_*$ .
- (b) Let  $f \in C^2([a, b])$  and  $x_*$  be a simple root of  $f$  in  $(a, b)$ , i.e.,  $f'(x_*) \neq 0$ . Apply (a) to show that there is a neighborhood of  $x_*$  where Newton method must converge if  $x_0$  is chosen in this neighborhood.
- (c) Suppose  $g \in C(\mathbb{R})$ . Show that if the iteration  $x_{n+1} = g(x_n)$  converges to  $x_* \in \mathbb{R}$  for some  $x_0 \in \mathbb{R}$ ,  $x_n \neq x_*$  for all  $n$  sufficiently large, and  $g$  is differentiable at  $x_*$ , then we must have  $|g'(x_*)| \leq 1$ . Find an example where  $g'(x_*) = 1$ .

**Question 3.**

In all of the following you will need to justify your answers.

- (a) State whether the following sequences converge sublinearly, linearly, superlinearly, or quadratically (give the fastest rate):

$$\left(\frac{1}{n^2}\right)_{n=1}^{\infty}, \quad \left(\frac{1}{2^{2^n}}\right)_{n=1}^{\infty}, \quad \left(\frac{1}{\sqrt{n}}\right)_{n=1}^{\infty}, \quad (e^{-n})_{n=1}^{\infty}, \quad \left(\frac{1}{n^n}\right)_{n=1}^{\infty}.$$

- (b) Let  $x_{n+1} = x_n + x_{n-1}$  with  $x_0 = 0$  and  $x_1 = 1$ . Let  $y_n := x_n/x_{n-1}$ . Is  $(y_n)_{n=1}^{\infty}$  convergent? If so, is the convergence sublinear, linear, superlinear, or quadratic?
- (c) Design an algorithm for computing the reciprocal of a positive real number  $a > 0$  that requires only addition and multiplication. For what values of  $x_0$  do the algorithm converges? Apply your algorithm to find the decimal expansion of  $1/12$  to 10 decimal digits of accuracy starting from  $x_0 = 0.1$  and  $x_0 = 1$ . Discuss your results.
- (d) Find the number of roots of  $2x = 3 \sin x$  in  $\mathbb{R}$  and determine the largest root to three decimal accuracy.